Jiahao (Eric) Zhang

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Education

Ph.D. in Mathematics, University of Washington *Expected May 2027*
Advisor: James Zhang

Bachelor of Science in Mathematics, Washington and Lee University *2020*
Summa cum laude, Minor in East Asian Studies

Budapest Semesters in Mathematics *2019-2020*
Magas Kitüntetéssel (high honor) Fall 2019

Research Interests

I work in noncommutative algebra and homological algebra, with a focus on developing tools to study ring-theoretic properties of simplicial algebras and to formulate a theory of twisted Calabi-Yau simplicial algebras.

Publications and Preprints

Introduction to Separable Simplicial Algebras
Joint work with X. Huang, G. Sun, and J. J. Zhang.
arXiv:2610.38229 (2026).

Moore–Postnikov Supcation of Simplicial Sets and Applications in Simplicial Algebras
Joint work with X. Huang, G. Sun, and J. J. Zhang.
arXiv:2610.40672 (2026).

Manuscripts in Preparation

Simplicial Picard group and simplicial Brauer group, in preparation.
Joint work with X. Huang, G. Sun, and J. J. Zhang.

Representations of simplicial winding algebra, in preparation.
Joint work with X. Huang, G. Sun, and J. J. Zhang.

Awards

Department of Mathematics Excellence in Teaching Award, University of Washington *2026*

Nominated for University of Washington Excellence in Teaching Award *2026*

Graduate Student Conference Presentation Awards, University of Washington *2026*

Invited Talks

2026 Fall Western Sectional Meeting, Tempe, AZ *Nov. 2026*
Special Session on Interactions in Noncommutative algebra, Geometry, and Representation Theory
Talk: *Separable simplicial algebras*

UC San Diego Algebra Seminar *Oct. 2026*
Talk: *Separable simplicial algebras*

2026 Fall Eastern Sectional Meeting, Washington, DC *Oct. 2026*
Special Session on Noncommutative Algebra
Talk: *Simplicial Morita equivalence and invariants*

2026 Spring Southeastern Sectional Meeting, Savannah, GA *Mar. 2026*
Special Session on Noncommutative Algebras, Quantum Groups, and Related Topics
Talk: *Truncation of simplicial sets with applications in simplicial algebras*

Joint Mathematics Meetings, Seattle, WA *Jan. 2025*
Special Session on Homological Interactions Between Commutative and Noncommutative Algebra
Talk: *Infinity categories of dim* ≤ 1

Teaching

Instructor of Record, University of Washington *2025*
Math 411: Introduction to Modern Algebra with Applications I

Teaching Certificate, University of Washington *2025*
Completed training in online and hybrid teaching

Teaching Assistant, University of Washington *2020-2026*
Math 112: Application of Calculus to Business and Economics
Math 124-126: Calculus with Analytic Geometry I, II, and III
Math 208: Matrix Algebra with Applications
Math 224: Advanced Multivariable Calculus
Math 300: Introduction to Mathematical Reasoning
Math 301: Elementary Number Theory
Math 327: Introductory Real Analysis I
Math 340: Abstract Linear Algebra
Math 404: Introduction to Modern Algebra (Galois theory)
Math 427: Complex Analysis I
Math 428: Complex Analysis II

Graduate Tutor, University of Washington *2020-2022, 2024*
Math Study Center, helped compile a preliminary Calculus Tutor Guide

Mentoring

Graduate Student Mentor, Washington Experimental Mathematics Lab *2023-2024*
University of Washington
Mentored two undergraduate researchers studying combinatorial, computational, and homological aspects of quadratic algebras under the direction of Be'eri Greenfeld.

Mentor, Washington Directed Reading Program *2020-2021*
University of Washington
Led a reading course on category theory for a senior undergraduate

Conferences, Workshops, and Seminars

Seattle Noncommutative Algebra Day, Seattle, WA *Scheduled 2026*

2026 Fall Western Sectional Meeting, Tempe, AZ *Scheduled 2026*

UCSD Algebra Seminar, La Jolla, CA *2026*

2026 Fall Eastern Sectional Meeting, Washington, DC *2026*

2026 Spring Southeastern Sectional Meeting, Savannah, GA *2026*

Seattle Noncommutative Algebra Conference, Seattle, WA *2025*

Revisiting Fundamental Problems Workshop, SLMath, CA *2025*
Infinite-Dimensional Division Algebras - Algebraicity and Freeness

Joint Mathematics Meetings, Seattle, WA *2025*

SLMath Summer Graduate School, St. Mary's College, CA *2024*
Multigraded and differential graded methods in commutative algebra

Introductory Workshop: Commutative Algebra, SLMath, CA *2024*

Introductory Workshop: Noncommutative Algebraic Geometry, SLMath, CA *2024*

Seattle Noncommutative Algebra Day, Seattle, WA *2023*

AMS Spring Western Sectional Meeting, California State University, Fresno, CA *2023*

AMS Spring Central Sectional Meeting, University of Cincinnati, Cincinnati, OH *2023*

Seattle Noncommutative Algebra Day, Seattle, WA *2022*

MSRI Summer Graduate School, University of Hawaii at Hilo, Hilo, HI *2022*
New Directions in Representation Theory